# Introduction to Machine Learning

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# Lecture 1: Introduction

Machine learning uses data to find patterns and make predictions. A model describes a family of possible patterns, and a learning algorithm uses data to select a model. The amount and quality of the data affect what can be learned.

## 1.1 Classification

In CLASSIFICATION, each example has a feature vector $x$ and a class label $y$. Given labeled examples, the goal is to predict the label of a new example. For instance, a lender might use an applicant's income and credit card balance to predict whether the applicant will default. The two coordinates describe a point; its label records the observed outcome.

A DECISION BOUNDARY separates regions assigned to different classes. A linear classifier uses a straight line in two dimensions, or a hyperplane in higher dimensions. Another approach is $k$-NEAREST NEIGHBORS: for a new point, find the $k$ closest training points and predict the majority label among them. With $k = 1$, the prediction is the label of the single nearest training point. Different models can draw very different boundaries from the same data.

A digital image can also be represented by a vector: list its pixel intensities as coordinates. An image with 16 pixels is a point in 16-dimensional space. A classifier can learn from labeled images of handwritten digits, then predict whether a new image is a seven.

## 1.2 Training, validation, and testing

The objective is to GENERALIZE: predict accurately on examples not used to fit the model. Split labeled examples into distinct sets before training.

1. Use the TRAINING SET to fit each candidate classifier's parameters.
2. Use the VALIDATION SET to compare trained classifiers and choose settings such as $k$. Do not use these examples to fit their parameters.
3. Use a separate TEST SET for a final evaluation of the selected classifier on new data. Test labels may be withheld by the evaluator.

The classification error on a set is the fraction of its examples assigned the wrong label. Training error measures fit to the examples used for learning; validation error helps select a model; test error assesses the final choice. Terminology can vary: sometimes "training data" means all available labeled data, so check which subset is meant.

For example, a 1-nearest-neighbor classifier classifies each training point correctly when its features identify it uniquely and its label is consistent. That does not imply good predictions on unseen points. Choosing $k$ only by training error would favor $k = 1$ even when a larger $k$ generalizes better.

### 1.3 Overfitting and underfitting

A HYPERPARAMETER controls the learning procedure or model family rather than being fitted as an ordinary model parameter. The number $k$ in nearest neighbors is one example. Small $k$ permits an intricate boundary that follows individual examples. Large $k$ averages over more neighbors and usually produces a smoother boundary.

OVERFITTING occurs when predictions on new data deteriorate because the classifier is too sensitive to noise, outliers, or accidental patterns in the training set. An OUTLIER is an atypical example, such as a wealthy borrower who nevertheless defaulted. A boundary that bends around every unusual point may fit the training data while predicting poorly elsewhere.

UNDERFITTING occurs when a model is too inflexible to capture important patterns. A very large neighborhood, for example, can smooth away real local differences. Validation error helps choose a compromise between these extremes. Repeatedly selecting models based on the same test results can itself favor a lucky model, so a final private test set can provide a more trustworthy assessment.

**Source.** Jonathan Shewchuk, Introduction to Machine Learning, Spring 2025, Lecture 1.